Now, the Earth does *not feel the average* of these gravitational pulls, because it is falling freely along its orbit.[11] (This is like the *Endurance*'s crew not feeling Gargantua's gravitational pull when they are in the *Endurance*, in its parking orbit above the black hole. They only feel centrifugal forces due to the *Endurance*'s rotation.) What the Earth *does* feel is the red-arrowed lunar pulls in the left half of Figure 4.8, with their average subtracted away; that is, it feels a stretch toward and away from the Moon, and a squeeze on its lateral sides (right half of Figure 4.8). This is qualitatively the same as around a black hole (Figure 4.7).

These felt forces stretch the ocean away from the Earth's surface on the faces toward and away from the Moon, producing high tides there. And the felt forces squeeze the oceans toward the Earth's surface on the Earth's lateral sides, producing low tides there. As the Earth turns on its axis, one full turn each twenty-four hours, we see two high tides and two low tides. This was Newton's explanation of ocean tides, aside from a slight complication: The Sun's tidal gravity also contributes to the tides. Its stretch and squeeze get added to the Moon's stretch and squeeze.

Because of their role in ocean tides, these gravitational squeezing and stretching forces—the forces the Earth *feels*—are called tidal forces. To extremely high accuracy, these tidal forces, computed using Newton's laws of gravity, are the same as we compute using Einstein's relativistic laws. They must be the same, since the relativistic laws and the Newtonian laws always make the same predictions when gravity is weak and objects move at speeds much slower than light.

In the relativistic description of the Moon's tides (Figure 4.9), the tidal forces are produced by blue tendex lines that squeeze the Earth's lateral sides and red tendex lines that stretch toward and away from the Moon. This is just like a black hole's tendex lines (Figure 4.7). The Moon's tendex lines are visual embodiments of the Moon's warping of